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INSTITUTE OF SCIENCE AND TECHNOLOGY
GODAWARI COLLEGE, ITAHARI

A Midterm Progress Report On
“NEXL: Elevate Your Learning”

Submitted To:
DEPARTMENT OF B.Sc. CSIT
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*In partial fulfillment of the requirements for the degree of Bachelor of Science in Computer
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Supervisor Recommendation

This is to clarify that this project prepared by Sachin Sharma, Prashiksha Shrestha, and Sidhant Dhungel, students of BSc. CSIT 2079 Batch, Godawari College entitled “**NEXL: Elevate Your Learning**”, is a web-based online learning platform developed to provide users with access to various educational courses, including business, technology, and skill-development programs. This project is carried out as partial fulfillment of the requirements for the B.Sc. CSIT degree.

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Mr. Dipesh Koirala

Supervisor

Acknowledgement

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Abstract

The NEXL:Elevate Your Learning platform is designed to enhance the learning experience by providing an interactive and accessible web-based educational system. Traditional learning environments often face challenges such as limited accessibility, lack of personalized learning paths, and reduced student engagement. NEXL aims to overcome these challenges by offering a centralized platform where learners can enroll in courses, access structured learning materials, participate in quizzes, track progress, and earn achievement badges.

The system is developed using the MERN stack (MongoDB, Express.js, React.js, Node.js), ensuring efficient data management, responsive user interfaces, and secure handling of user information. Key features of the platform include course management, user authentication, progress tracking, quiz and assessment modules, badge-based gamification, and administrative control for content management. NEXL provides a user-friendly, reliable, and scalable solution that supports both learners and educators, improving engagement, motivation, and overall learning effectiveness.

Contents

Supervisor Recommendation	i
Acknowledgement	ii
Abstract	iii
List of Abbreviations	viii
1 Introduction	1
1.1 Introduction	1
1.2 Problem Statement	1
1.3 Objectives	3
1.4 Scope and Limitations	4
1.5 Development Methodology	6
1.6 Report Organization	6
1.7 Expected Outcome	7
2 Background Study and Literature Review	9
2.1 Background Study	9
2.2 Literature Review	11
3 System Analysis	14
3.1 System Analysis	14
4 System Design	26
4.1 Algorithm Details and High-Level Design	26
4.2 High-Level Design	29
5 Implementation and Testing	31
5.1 Implementation	31
5.2 Testing and Verification	32
5.3 Result Analysis	32
5.4 Development Roadmap and Task Status	32
6 Conclusion and Future Recommendations	34

6.1 Conclusion	34
6.2 Future Recommendations	34
REFERENCES	38

List of Figures

1.1	Water Flow Diagram	6
3.1	Use Case Diagram	16
3.2	Object Diagram	22
3.3	Sequence Diagram	23
3.4	Activity Diagram	25
4.1	Flowchart	27
4.2	Block Diagram	30
6.1	Login page	35
6.2	Main Page	36
6.3	API TESTING	37
6.4	API TESTING	37

List of Tables

2.1	Comparison of Existing Solutions vs. NEXL Requirements	13
3.1	Estimated Development Cost of the NEXL Platform	20

List of Abbreviations

AI	– Artificial Intelligence
API	– Application Programming Interface
BART	– Bidirectional and Auto-Regressive Transformers
CRUD	– Create, Read, Update, Delete
HTTP	– HyperText Transfer Protocol
HTTPS	– HyperText Transfer Protocol Secure
JWT	– JSON Web Token
LMS	– Learning Management System
MERN	– MongoDB, Express.js, React.js, Node.js
REST	– Representational State Transfer
SDK	– Software Development Kit
SDLC	– Software Development Life Cycle
UML	– Unified Modeling Language
URL	– Uniform Resource Locator
VPS	– Virtual Private Server

1 Introduction

1.1 Introduction

With the rapid growth of digital technology and internet access, online learning has become an essential part of modern education. E-learning platforms enable students to access educational content anytime using computers, tablets, or smartphones. The global e-learning market has expanded as institutions, instructors, and organizations increasingly adopt digital solutions for courses, assessments, and student interaction. This shift creates learning environments where students actively engage in educational processes through instructional strategies integrated with ICT tools [1].

NEXL: Elevate Your Learning is an online platform connecting students and independent educators in a structured digital environment. Like traditional Learning Management Systems (LMS), NEXL provides tools to manage learning and monitor performance for both students and instructors. Guests can browse courses, view descriptions, and explore categories before registering.

Registered students can enroll in courses, watch video lectures, access notes and PDFs, follow structured lessons, take quizzes, submit assignments, participate in discussions, and rate or review courses. Personalized course recommendations and smart search help learners find relevant content efficiently.

Independent instructors can create and manage courses, upload materials, set pricing, offer previews or discounts, and design quizzes and assignments. Research shows that effective LMS adoption depends on providing tools that meet the specific needs of each user group [2].

Built with modern web technologies, **NEXL** delivers a responsive, accessible, and high-performance experience. The platform emphasizes affordability and practical learning, supporting professional development for instructors and reliable access across devices. By empowering both learners and educators, NEXL aims to advance digital education in Nepal.

1.2 Problem Statement

The rapid growth of online education has created new opportunities for flexible and accessible learning worldwide. In Nepal, however, the development of effective and sustainable Learning Management Systems (LMS) is constrained by several practical, financial, and technological challenges. While digital platforms have the potential to transform education delivery, issues related to payment systems, user trust, and platform capabilities continue to hinder their widespread adoption.

Research indicates that despite increasing internet penetration and ongoing digital transformation, the adoption of electronic payment systems in Nepal remains limited due to concerns related to accessibility, usability, and user trust [3,4]. These challenges directly impact the commercialization and long-term sustainability of online learning platforms.

The major problems addressed by this project are categorized into three key dimensions:

Financial and Payment Barriers

- Limited accessibility of international payment systems: Global payment gateways such as PayPal and Stripe do not currently support outgoing payments to Nepal, creating significant barriers for local educators seeking to monetize their courses [5].
- Transaction reliability and usability challenges: Studies highlight that issues such as system unreliability, complex interfaces, and inconsistent performance reduce adoption of digital payment platforms among Nepali users [6].
- Low trust in digital transactions: User perception studies show that security concerns, lack of trust, and accessibility limitations significantly affect the adoption of online payment systems in Nepal [4, 7].

Monetization and Pricing Limitations

- Limited monetization flexibility: Many locally developed platforms lack advanced pricing mechanisms such as discount codes, course previews, and dynamic pricing models, which are essential for attracting and retaining students in a competitive market.
- Insufficient instructor support tools: Effective LMS adoption requires providing instructors with tools for pricing management, performance tracking, and content structuring. Research emphasizes the importance of aligning platform features with the specific needs of each stakeholder group to ensure successful implementation [8].

Feature and Engagement Gaps

- **Lack of advanced interactive features:** Modern LMS platforms increasingly rely on real-time communication, AI-personalized learning paths, and intelligent course recommendations. However, many systems available in Nepal lack these capabilities, reducing student engagement and overall learning effectiveness [2].
- Usability and infrastructure challenges: Studies on digital platforms in Nepal highlight poor user interface design, inconsistent system behavior, and infrastructure limitations as key factors that negatively affect user experience and platform adoption [6].

These interconnected challenges discourage educators from creating high-quality digital content, reduce student willingness to invest in paid courses, and limit the growth of a sustainable online education ecosystem in Nepal [9]. There is therefore a strong need for a modern, locally-tailored LMS that integrates accessible payment solutions, flexible monetization strategies, and AI-enhanced interactive features.

1.3 Objectives

The primary objective of this project is to design and develop **NEXL**, a web-based Learning Management System (LMS) that improves accessibility, interactivity, and personalization in online education while addressing Nepal-specific financial and infrastructural challenges.

The specific objectives of the project are as follows:

Localization and Payment Integration

- To integrate widely used Nepali digital payment systems such as eSewa, and Fonepay in order to facilitate convenient and accessible transactions for users.
- To design and implement flexible pricing mechanisms, including discount and coupon systems, dynamic pricing options, and course preview features, to enhance user engagement and build trust among learners.

Instructor Empowerment

- To develop comprehensive tools for instructors that support course creation, content management, pricing control, and basic analytics for monitoring student performance and engagement.

1.3.1 AI-Enhanced Learning Experience

- To incorporate AI-based functionalities such as automated text summarization of learning materials and personalized course recommendations to improve learning efficiency and content discovery.

Interactive Assessment and Engagement

- To design and implement assessment modules including quizzes and assignments with automated evaluation and timely feedback mechanisms.
- To introduce gamification elements such as performance-based digital badges and achievement indicators to enhance student motivation and participation.

Communication and Accessibility

- To implement communication features including discussion forums, group interaction, and one-on-one session booking to support collaborative learning.
- To ensure system usability through responsive design, intuitive navigation, and compatibility across multiple devices, considering the technological context of Nepal.

1.4 Scope and Limitations

1.4.1 Scope

The NEXL platform is a web-based Learning Management System designed for the Nepali educational context. The scope of the system includes the following components:

- Core Platform: Development of a responsive web application using React.js, Node.js, Express.js, and MongoDB, with consideration for usability under low to moderate bandwidth conditions.
- Payment Infrastructure: Integration of widely used Nepali digital payment systems—eSewa and —to enable convenient course enrollment. The system supports:
 - Coupon and discount code mechanisms (percentage-based and fixed-value)
 - Flexible pricing configuration by instructors
 - Free course preview or demo access
- User Management: Implementation of three primary user roles with role-based access control:
 - Student: Course enrollment (free and paid), video streaming, participation in quizzes and assignments, progress tracking, and access to AI-generated content summaries.
 - Instructor: Course creation with multimedia content, pricing management, revenue tracking, and analytics on student engagement and performance.
 - Admin: User management, course approval workflows, and platform activity monitoring.
- AI-Enhanced Features:
 - Abstractive text summarization of course materials using transformer-based models (e.g., facebook/bart-large-cnn) accessed via the Hugging Face Inference API.

- AI-assisted course metadata generation (descriptions, categories, and tags) from course titles using language model inference.
- **Interactive Communication:** Real-time communication features including group chat and one-on-one tutoring session booking to support interaction between students and instructors, powered by the Stream SDK.
- **Multimedia Delivery:** Support for video-based learning content, document uploads (PDF), and live class capabilities via Cloudinary (media storage) and Stream Video API (live streaming).

1.4.2 Limitations

- **Platform Scope:** The system is limited to a web-based application. Native mobile applications (iOS/Android) are not developed, although the platform is accessible via mobile browsers.
- **Dependency on External Services:** AI functionalities and multimedia services rely on third-party APIs (Hugging Face, Stream SDK, Cloudinary). System performance may be affected by API availability, latency, and free-tier usage limits.
- **Live Streaming Constraints:** Live session features are implemented using the Stream Video managed service rather than a custom-built infrastructure, which limits advanced customization and introduces external dependency.
- **Internet Dependency:** The system requires an active internet connection for full functionality. Offline access and content synchronization are not supported.
- **Payment Coverage:** The platform focuses on Nepali digital payment systems (eSewa and) and does not include international payment gateways, limiting accessibility for users outside Nepal.
- **Assessment Security:** Automated evaluation is implemented for quizzes and assignments; however, advanced proctoring and anti-cheating mechanisms are not included in the current version.
- **Recommendation System:** The current implementation provides AI-assisted metadata and keyword-based course discovery. A full semantic recommendation engine using vector embeddings is planned for a future release.

1.5 Development Methodology

The development of the NEXL platform follows a systematic Software Development Life Cycle (SDLC) based on the Water Fall Model. This approach ensures the system remains adaptable to the specific needs of the Nepali educational market through continuous refinement and feature progression across functional modules such as AI assistants and payment processing. This model allow us to deliver a base product incrementally while refining requirements based on stakeholder feedback in each cycle.

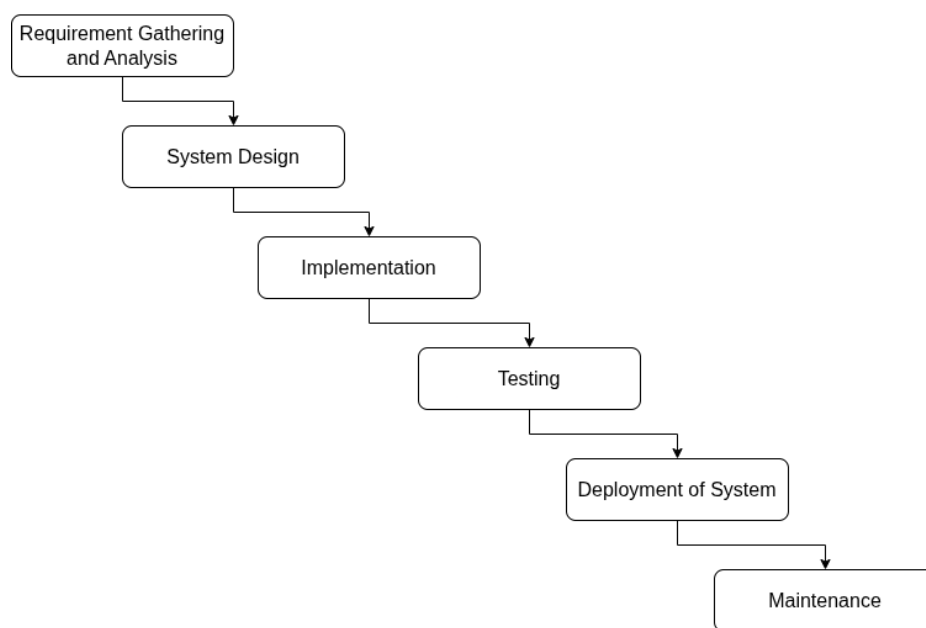


Figure 1.1 Water Flow Diagram

1.6 Report Organization

This project report is organized into six chapters to document the complete development lifecycle of the NEXL platform:

Chapter 1: Introduction provides the background of the project, problem statement, objectives, scope and limitations, development methodology, and report organization.

Chapter 2: Background Study and Literature Review discusses fundamental theories related to Learning Management Systems, e-learning in developing countries, artificial intelligence in education, and reviews existing global and local platforms with gap analysis.

Chapter 3: System Analysis presents requirement analysis (functional and non-functional), feasibility analysis (technical, operational, economic, and schedule), and object-oriented analysis using UML diagrams including use case, class, sequence, state, and activity diagrams.

Chapter 4: System Design describes the system architecture, database design, interface

design, and detailed algorithms including course recommendation, content summarization, and search algorithms.

Chapter 5: Implementation and Testing covers the technological stack, implementation details of modules, unit testing, system testing, and result analysis.

Chapter 6: Conclusion and Future Recommendations summarizes the work completed, achievements, and suggests future enhancements for the platform.

The report also includes references, bibliography, and appendices containing screenshots, source code snippets, and supervisor visit logs.

1.7 Expected Outcome

Upon successful completion of the project, NEXL: Elevate Your Learning will deliver a fully functional, full-stack web-based Learning Management System designed to support modern digital education. The platform aims to provide an accessible and flexible learning environment for both instructors and students.

The expected outcomes of the project are as follows:

- **A complete and deployable web application** that enables independent teachers and tutors to:
 - Create and manage courses using multiple content formats such as video lectures, handwritten or scanned notes, and PDF documents.
 - Set flexible pricing structures and apply discount coupons for courses.
 - Provide optional preview or demo videos to introduce course content to potential students.
 - Communicate with students through real-time chat groups and announcement channels.
 - Create assignments and quizzes with automated evaluation features.
 - Award performance-based digital badges to motivate students based on quiz performance, assignment completion, and overall participation.
- **An intuitive student-oriented learning platform** that allows users to:
 - Browse, search, and discover available courses efficiently.
 - Receive basic personalized course recommendations based on their interests and previous interactions.
 - Securely purchase and enroll in courses using integrated payment systems.

- Access enrolled course materials, participate in discussions, complete assessments, and earn digital badges.
- Provide ratings and reviews to help maintain course quality and improve platform credibility.
- **Improved accessibility and affordability** by supporting flexible learning resources such as handwritten notes, offering discount mechanisms, and minimizing platform costs to make education more accessible to students from diverse economic backgrounds.
- **Contribution to the digital education ecosystem** by empowering independent educators to share knowledge, generate sustainable income, and provide students with engaging and interactive learning experiences.

The final system will be deployed on a cloud hosting platform and will be fully tested, documented, and demonstrated during the final project defense. The developed system will serve as a scalable prototype that can be further expanded in the future with additional features such as mobile application support, advanced recommendation systems, and extended payment options.

Overall, the NEXL platform is expected to provide a practical, user-friendly, and scalable solution for digital learning, addressing key challenges related to accessibility, content flexibility, interaction, and motivation within the growing e-learning landscape.

2 Background Study and Literature Review

2.1 Background Study

2.1.1 Learning Management Systems (LMS)

A Learning Management System (LMS) is a web-based platform designed to deliver, manage, and evaluate educational content in a structured manner. It acts as a centralized system where instructors can upload materials, track student progress, and conduct assessments. LMS platforms are widely used in schools, universities, and corporate training environments due to their flexibility and accessibility. They support both synchronous and asynchronous learning, making education more accessible regardless of location. Key functionalities include content delivery, user management, assessment tools, communication features, and analytics for performance tracking [1]. Modern LMS platforms are built using scalable architectures such as service-oriented design, RESTful APIs, and microservices, enabling seamless integration with third-party services. These systems also support mobile learning, ensuring accessibility on smartphones and tablets.

Core components include:

- **Content Delivery:** Provides videos, documents, and multimedia learning resources
- **User Management:** Handles authentication, authorization, and role-based access
- **Assessment:** Supports quizzes, assignments, grading, and feedback systems
- **Communication:** Includes discussion forums, messaging, and live chat features
- **Analytics:** Tracks learner progress and performance metrics

2.1.2 E-Learning in Developing Countries

E-learning has emerged as a powerful solution for expanding access to education in developing countries. However, several challenges limit its effectiveness, including poor internet infrastructure, low digital literacy, and financial constraints. In countries like Nepal, reliable internet access is not always available, especially in rural areas, making it difficult to use high-bandwidth platforms. Therefore, e-learning systems must be optimized for low bandwidth and mobile devices. Additionally, users often rely on smartphones rather than computers, which influences system design. Financial limitations also affect the adoption of paid platforms, emphasizing the need for affordable solutions. During the COVID-19 pandemic, the rapid shift to online learning exposed weaknesses such as lack of localization,

limited interactivity, and absence of integrated payment systems [5,9]. These challenges highlight the importance of developing context-aware LMS platforms tailored to local needs.

2.1.3 Artificial Intelligence in Education

Artificial Intelligence (AI) is transforming modern education by improving efficiency, personalization, and learner engagement. AI-powered systems can analyze user behavior and adapt content based on individual learning patterns. This enables personalized learning experiences where students receive recommendations suited to their progress and interests. AI also automates repetitive tasks such as grading and content organization, reducing the workload for instructors. In addition, intelligent tutoring systems can provide real-time feedback and guidance. The integration of AI into LMS platforms enhances both teaching and learning processes. As a result, AI-driven education systems are becoming increasingly popular in both academic and professional environments.

Automatic Text Summarization

Automatic text summarization is an AI technique used to generate concise summaries from large volumes of text. Modern approaches use transformer-based models such as BERT and BART, which rely on encoder-decoder architectures and attention mechanisms. These models understand context and semantics, producing accurate and meaningful summaries. Summarization is particularly useful in e-learning, where students need to quickly grasp key concepts from lengthy materials. It reduces study time while improving comprehension and retention. This feature is especially beneficial for mobile users who prefer shorter, digestible content [10].

Recommendation Systems

Recommendation systems enhance user experience by suggesting relevant learning materials based on user preferences and behavior. Content-based filtering techniques use embeddings such as E5 and MPNet to represent textual data in vector form. Similarity measures like cosine similarity are then used to match users with appropriate content. Advanced systems utilize vector databases such as Faiss or Pinecone for efficient storage and retrieval of embeddings. These systems improve engagement by guiding learners toward relevant courses and materials. As a result, recommendation engines play a crucial role in modern AI-powered LMS platforms [2].

2.1.4 Digital Payment Integration in E-Education

Digital payment integration is essential for enabling monetization in online education platforms. It allows instructors to sell courses and users to make secure transactions. In Nepal, local payment gateways such as eSewa provide affordable transaction options compared to international services like PayPal or Stripe. These local solutions typically charge lower transaction fees, making them suitable for small-scale platforms [5]. Integration involves APIs for payment initiation, verification, and status tracking. Security is a critical aspect, requiring encryption protocols such as SSL/TLS and mechanisms like HMAC for data integrity. Proper transaction handling ensures reliability and user trust in the system.

Key implementation aspects:

- **API Integration:** Handles payment requests and verification processes
- **Security:** Ensures safe transactions using encryption and authentication
- **Transaction Handling:** Manages payment states such as success, failure, and pending

2.2 Literature Review

2.2.1 Global Learning Management Systems

Moodle

Moodle is one of the most widely used open-source LMS platforms, known for its flexibility and extensive feature set. It supports course management, assessments, and collaborative learning tools. Being open-source, it allows customization according to institutional needs. However, its setup and maintenance require technical expertise, which can be a barrier in resource-limited environments. Additionally, performance optimization and scalability can be challenging without proper infrastructure [1].

Canvas and Blackboard

Canvas and Blackboard are commercial LMS platforms offering advanced features such as analytics, integrations, and cloud hosting. They provide a user-friendly interface and robust support services. However, these platforms are expensive and often not affordable for institutions in developing countries. They also lack integration with local payment systems, limiting their usability in regions like Nepal [10].

Google Classroom

Google Classroom is a free and easy-to-use platform widely adopted during the COVID-19 pandemic. It integrates well with other Google services and supports basic classroom management features. However, it lacks advanced functionalities such as monetization, deep customization, and AI-driven analytics. This makes it less suitable for scalable and feature-rich educational platforms [9].

2.2.2 Online Education in Nepal

In Nepal, online education is still developing and largely depends on general-purpose tools such as Zoom, Google Meet, and social media platforms. During the pandemic, approximately 68% of institutions used platforms like Facebook and YouTube for teaching. These tools lack structured LMS capabilities such as assessment management and progress tracking. Local platforms such as Mero School and Gurukul primarily focus on K-12 education and offer limited flexibility. They also lack monetization features and AI-based enhancements, which are essential for modern e-learning systems [5, 9].

2.2.3 AI-Enhanced Educational Platforms

AI-enhanced platforms have demonstrated significant improvements in learning outcomes and engagement. Studies indicate that automatic summarization can reduce study time by up to 40%, allowing students to learn more efficiently. Similarly, recommendation systems can increase user engagement by approximately 35% by providing relevant content. These technologies help create personalized learning environments, making education more effective and interactive. As AI continues to evolve, its role in education is expected to grow further [2, 10].

2.2.4 Gap Analysis

Existing systems show several limitations when applied to the Nepali context. International LMS platforms lack local payment integration and are often expensive, while local platforms lack advanced features such as AI and scalability. There is a clear gap between available solutions and the requirements of a modern, localized LMS. This highlights the need for a system like NEXL, which integrates AI features, supports local payments, and provides flexible monetization options tailored to Nepal.

Table 2.1 Comparison of Existing Solutions vs. NEXL Requirements

Feature	International LMS	Local Platforms	NEXL (Proposed)
Local Payment Integration	Not Available	Partial	Full (eSewa)
AI Summarization	Premium/Add-on	Not Available	Integrated
Recommendation Engine	Basic	Not Available	Vector-based
Instructor Monetization	High Commission	Fixed Pricing	Flexible
Real-time Interaction	Limited	Basic	Integrated

These gaps justify the development of NEXL as a localized, AI-enabled LMS tailored to the needs of Nepal.

3 System Analysis

3.1 System Analysis

3.1.1 Requirement Analysis

Functional Requirements (Illustrated using use case diagram/use case descriptions)

The functional requirements define the specific behaviors and functions that the NEXL platform must perform to meet the needs of students, instructors, and administrators. These requirements are categorized based on user roles and system modules.

User Management Module:

- **Authentication:** Secure user registration and login with JWT token-based authentication, supporting email/password and OAuth (Google/GitHub) providers.
- **Profile Management:** Users can create and update profiles, upload profile pictures, and manage notification preferences.
- **Role-Based Access Control:** Three distinct roles (Student, Instructor, Admin) with specific permissions enforced through middleware.

Course Management Module:

- **Course Creation:** Instructors can create courses with titles, descriptions, syllabi, and pricing in NPR (Nepali Rupees).
- **Content Upload:** Support for video files (MP4), documents (PDF), and images with Cloudinary integration.
- **Course Catalog:** Students can browse, search, and filter courses by category, price, and rating.
- **Enrollment Management:** Automated enrollment upon payment verification with access control to course materials.

Payment and Monetization Module:

- **Local Payment Integration:** Seamless integration with eSewa APIs for NPR transactions.

- **Coupon System:** Instructors can generate discount codes with percentage or fixed amount reductions.
- **Revenue Tracking:** Dashboard for instructors to view earnings, enrollments, and transaction history.

AI-Enhanced Learning Module:

- **Content Summarization:** Automatic generation of text summaries from video transcripts using transformer models (BART/DistilBART).
- **Recommendation Engine:** Personalized course suggestions based on user behavior using vector embeddings (E5 model) and cosine similarity.

Interactive Features:

- **Real-time Communication:** Stream Video SDK-based group chat and one-on-one messaging between students and instructors.
- **Assessment System:** Quiz creation with multiple-choice questions, automated grading, and instant feedback.
- **Gamification:** Digital badges awarded for course completion, high quiz scores, and participation.

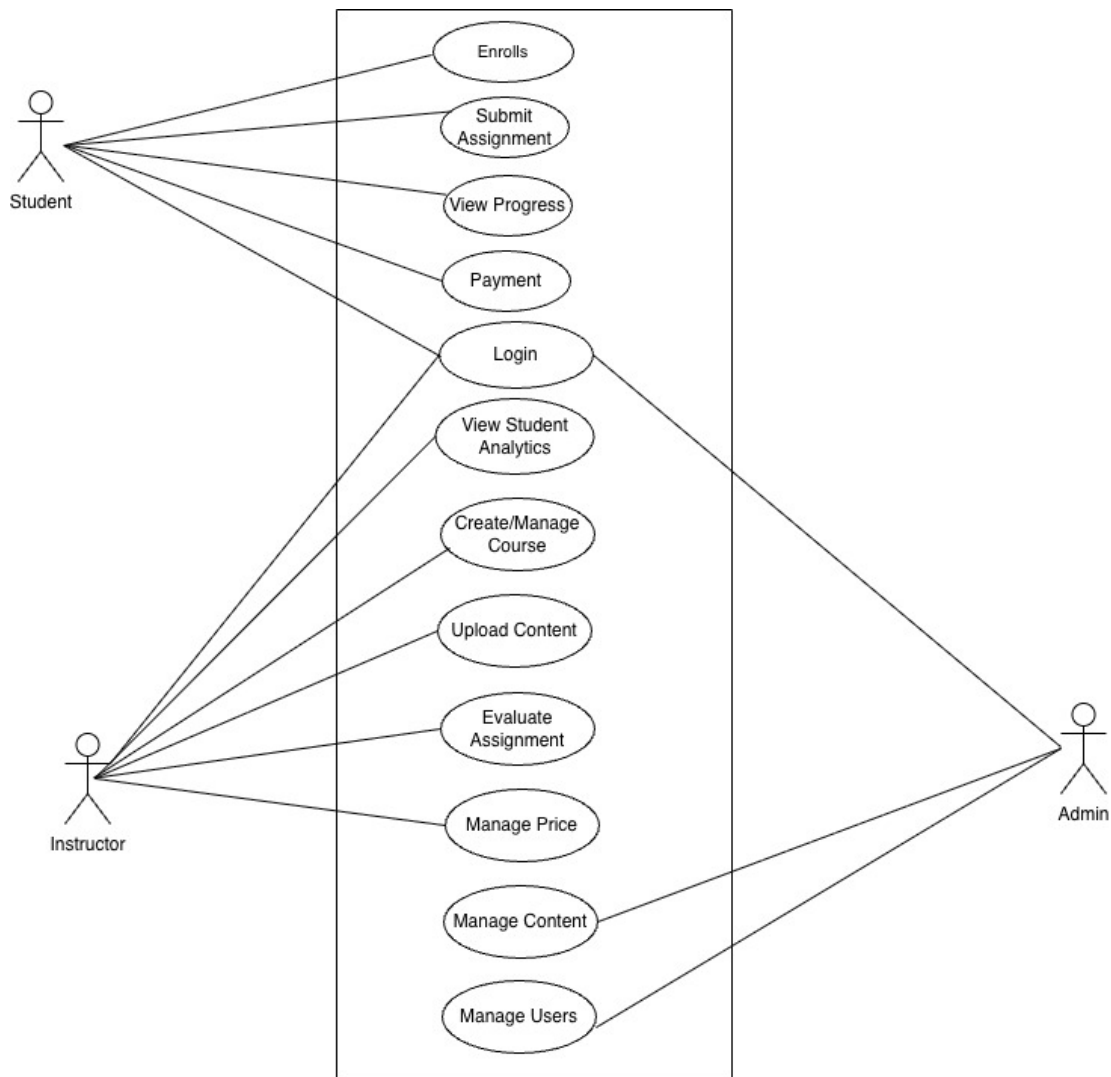


Figure 3.1 Use Case Diagram

Use Case Description: The use case diagram in Figure 3.1 illustrates the primary actors and their interactions with the NEXL system:

- **Student:** Can browse courses, enroll (free or paid), view content, take quizzes, submit assignments, participate in discussions, and receive AI-generated summaries.
- **Instructor:** Can create/manage courses, upload multimedia content, set pricing and coupons, evaluate assignments, view analytics, and communicate with students.
- **Admin:** Manages user accounts, approves published courses, monitors system health, and handles payment disputes.

Non Functional Requirements

Non-functional requirements specify the quality attributes, performance standards, and constraints under which the system must operate.

Performance Requirements:

- The system shall load the landing page within 3 seconds on a 5 Mbps internet connection (standard Nepali broadband).
- Video streaming shall support adaptive bitrate switching (480p, 720p, 1080p) to accommodate varying bandwidth conditions.
- The recommendation engine shall return results within 500ms for a catalog of up to 1,000 courses.

Security Requirements:

- **Authentication:** JWT tokens with 24-hour expiration and refresh token rotation.
- **Data Protection:** HTTPS/TLS 1.3 encryption for all data transmission; bcrypt hashing for passwords with salt rounds ≥ 10 .
- **Payment Security:** HMAC-SHA256 signature verification for payment gateway webhooks; no storage of sensitive banking credentials.
- **Input Validation:** XSS and SQL injection prevention through parameterized queries and input sanitization.

Usability and Accessibility:

- Responsive design supporting viewports from 320px (mobile) to 1920px (desktop).
- WCAG 2.1 Level AA compliance for accessibility (alt text for images, keyboard navigation, ARIA labels).
- Intuitive navigation with maximum 3-click depth to reach any course content.

Scalability and Reliability:

- Horizontal scaling capability through stateless REST API architecture.
- 99.5% uptime SLA with automated database backups every 24 hours.
- Support for concurrent users: minimum 500 simultaneous video streams.

3.1.2 Feasibility Analysis

Technical Feasibility

The NEXL platform is technically feasible within the scope of the B.Sc. CSIT curriculum and available open-source technologies.

Technology Stack Viability:

- **Frontend:** React.js 18+ with Vite build tool provides component-based architecture and efficient hot module replacement for development.
- **Backend:** Node.js with Express.js offers non-blocking I/O suitable for real-time chat and concurrent video streaming.
- **Database:** MongoDB Atlas provides flexible schema design for varying course structures and automatic sharding for scalability.
- **AI Integration:** Hugging Face Inference API with DistilBART (400 MB) models is feasible on a standard render free tier hosting
- **Payment APIs:** eSewa provides well-documented REST APIs with SDKs for Node.js, supporting sandbox environments for testing.

Infrastructure Requirements:

- **Development:** Standard laptop with 8 GB RAM, Node.js runtime, and MongoDB Community Edition.
- **Production:** React on Vercel And Node and Express on Render.
- **Media Storage:** Cloudinary free tier (25 GB) is sufficient for initial deployment.

Risk Assessment:

- **Low Risk:** MERN stack technologies are mature with extensive community support.
- **Medium Risk:** AI model inference latency; mitigated by caching frequent summaries and using smaller distilled models.
- **Mitigation:** Fallback to rule-based recommendations if vector search fails; manual payment verification as backup for API failures.

Operational Feasibility

Operational feasibility assesses whether the system aligns with existing organizational workflows and user capabilities.

User Acceptance:

- **Students:** Familiarity with e-learning platforms post-COVID-19; smartphone-first design accommodates primary device usage in Nepal (78% mobile penetration) [5].
- **Instructors:** Minimal technical training required due to intuitive dashboard design similar to YouTube Studio or Google Classroom.
- **Administrators:** Web-based admin panel with role-based access eliminates the need for specialized software installation.

Organizational Integration:

- The platform operates independently of existing institutional systems, requiring no integration with legacy university ERP systems.
- Content migration support: instructors can bulk-upload PDFs and MP4 files via a drag-and-drop interface.
- Support for both synchronous (live streaming) and asynchronous (recorded) learning modes accommodates varying internet reliability.

Maintenance Plan:

- **Content Moderation:** Admin approval workflow for courses before public visibility ensures quality control.
- **Update Cycle:** Monthly updates for security patches; quarterly feature updates based on user feedback.
- **Support System:** Built-in ticketing system for technical issues with a 48-hour SLA for critical payment-related bugs.

Economic Feasibility

Economic feasibility demonstrates that the project is financially viable with minimal upfront investment and sustainable operational costs.

Development Costs (One-time):

- **Software Licenses:** \$0 — Open-source stack (React, Node.js, MongoDB, Express).
- **Development Hardware:** \$0 — Existing personal computers and laptops.
- **AI API:** \$0 — Hugging Face Inference API free tier.
- **Internet/Utilities:** NPR 0 (development phase only).

Deployment Costs (Monthly Recurring):

- **Database:** \$0 — MongoDB Atlas M0 free tier sufficient for initial 1,000 users.
- **Media Storage:** \$0–\$25/month (Cloudinary free tier initially, scaling at 50 GB).
- **Domain & SSL:** NPR 0/year (.com.np domain + Let’s Encrypt SSL).

Revenue Model:

- **Commission Structure:** 5–10% platform fee per transaction.
- **Break-even Analysis:** With 50 courses sold monthly at an average of NPR 1,000, the platform generates NPR 50,000 revenue; a platform fee of NPR 2,500–5,000 covers hosting costs.
- **Teacher Incentive:** Higher profit margins (90–95%) encourage content creation compared to Udemy (37%) or Skillshare (30%).

Cost-Benefit Analysis: Total initial investment: NPR 0 (\$0 USD). With a projected monthly transaction volume of NPR 100,000+ by month 6, the system achieves a positive ROI within the first semester of operation.

Table 3.1 Estimated Development Cost of the NEXL Platform

Cost Component	Amount (NPR)	Remarks
Software & Development Tools	0	Open-source (Node.js, React, MongoDB)
Hardware	0	Personal PC
AI Inference API	0	Hugging Face Free-Tier
Internet & Connectivity	0	Development Phase (Pre-production)
Vercel and Render Hosting	0	DigitalOcean / Render (Trial/Free Tier)
Maintenance	0	Periodic student-led updates
Total Cost	0	Economically viable for local adoption

Schedule

The project timeline spans 16 weeks (one academic semester) following an Water Fall methodology

3.1.3 Analysis

The system analysis follows the **Object Oriented Approach** utilizing UML (Unified Modeling Language) diagrams to model the static structure, dynamic behavior, and business processes of the NEXL platform.

Object Modelling using Object Diagrams

Object modeling represents the instances of classes and their relationships at a specific point in time, illustrating the runtime state of the system. **Object Diagram (Instance Level):** An object diagram for a specific enrollment scenario is shown in Figure 3.2, demonstrating the runtime state when a student accesses a course:

- **sachin:Student** with attribute values (name="Sachin Sharma", email="sachin@email.com", enrolledCourses=[webDev, dataScience], badges=[fastLearner])
- **prashiksha:Instructor** with attribute values (name="Prashiksha Shrestha", coursesCreated=[webDev], revenue=45000)
- **webDev:Course** with state (title="Complete Web Development", price=1500, discountPrice=1200, status=PUBLISHED, category="Technology")
- **enroll1:Enrollment** linking sachin and webDev with (enrolledDate="2025-03-15", paymentStatus="COMPLETED", progressPercentage=45%)
- **payment1:Payment** with (amount=1200, gateway="eSewa", transactionId="TXN123456", status="VERIFIED")

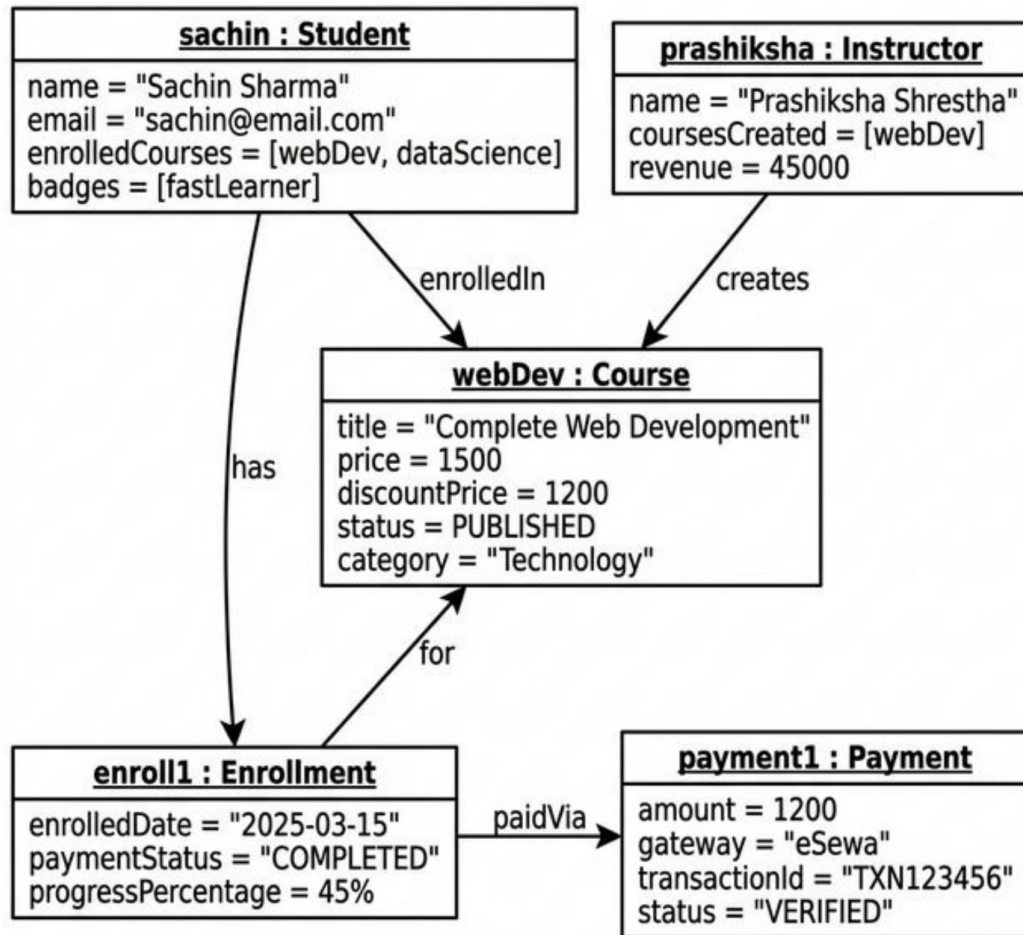


Figure 3.2 Object Diagram

This demonstrates the runtime state of the system when a specific student accesses a course, including the relationships between objects and their current attribute values.

Dynamic Modelling using Sequence Diagrams

Dynamic modeling illustrates the time-dependent behavior of objects and their interactions through message passing over time.

Sequence Diagram: Course Enrollment with Payment The sequence diagram in Figure 3.3 illustrates the interaction flow for a student enrolling in a paid course:

1. **Actor:** Student initiates enrollment via Frontend.
2. **Step 1:** Frontend sends POST /enroll request to Backend API with courseId and userId.
3. **Step 2:** Backend validates JWT token and checks if course exists and is published.

4. **Step 3:** Backend checks Enrollment collection to prevent duplicate enrollment.
5. **Step 4:** If paid course, Backend initiates payment with selected Gateway (eSewa/).
6. **Step 5:** Payment Gateway returns redirect URL; Frontend redirects user to payment page.
7. **Step 6:** User completes payment on Gateway; Gateway sends webhook to Backend.
8. **Step 7:** Backend verifies payment signature and updates Enrollment status to "COMPLETED".
9. **Step 8:** Backend grants access to Course Content; Frontend displays success message.

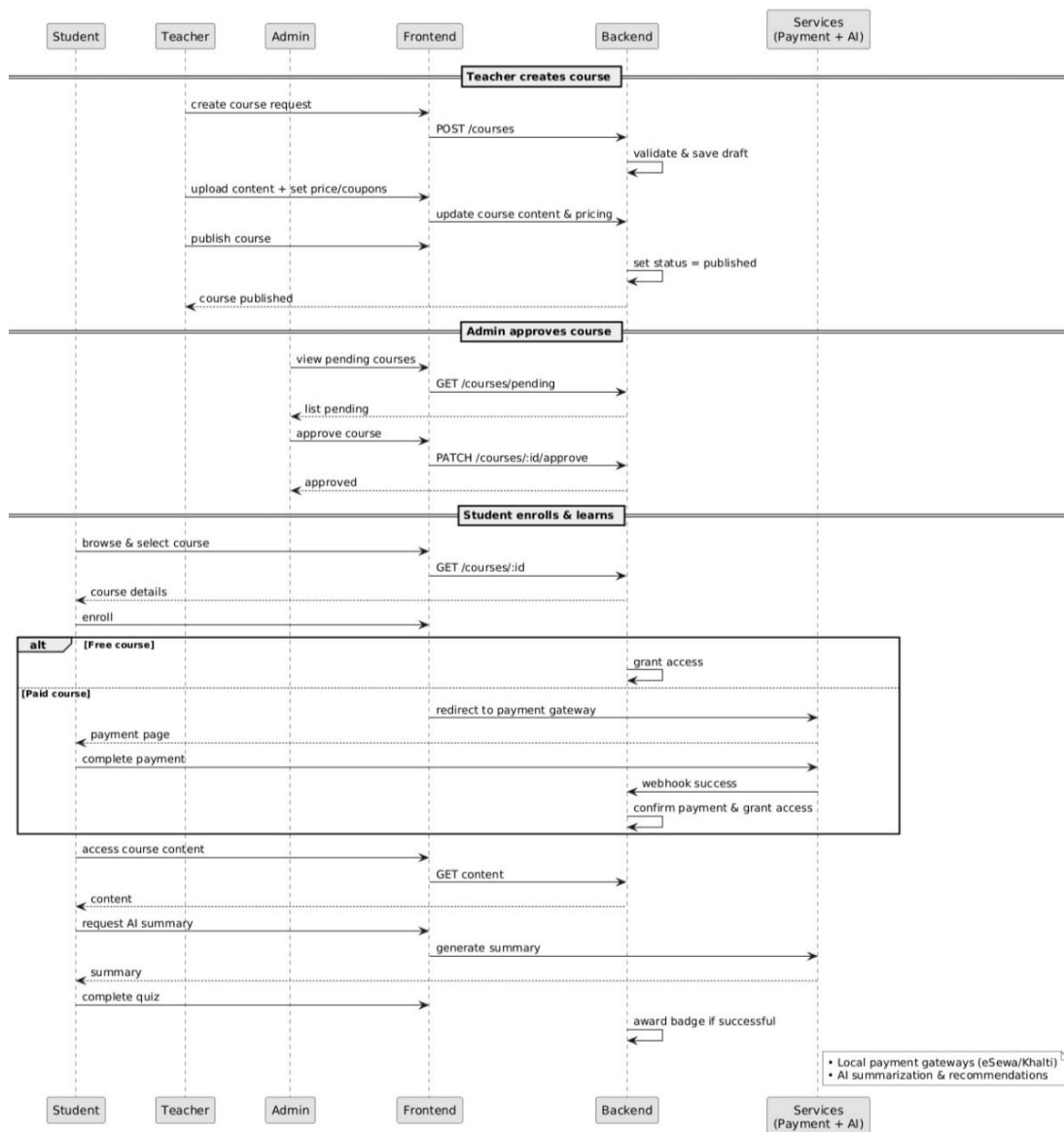


Figure 3.3 Sequence Diagram

Process Modelling using Activity Diagrams

Process modeling using Activity Diagrams illustrates the workflow and business logic of the course enrollment process, including decision points and parallel activities.

Activity Diagram: Course Enrollment and Payment Process

The activity diagram in Figure 3.4 models the complete workflow from browsing to content access:

Swimlanes (Partitions):

1. **Student:** Actions performed by the learner.
2. **System:** Backend processing and validation.
3. **Payment Gateway:** External eSewa/ services.
4. **Instructor:** Notifications and revenue updates.

Flow Description:

1. **Initial Node:** Student browses course catalog.
2. **Decision:** Is user authenticated?
No → Redirect to Login Activity → Return to Browse.
Yes → View Course Details.
3. **Decision:** Is course free?
Yes → Direct Enrollment Activity → Grant Access.
No → Check/Apply Coupon Activity → Calculate Final Price.
4. **Fork (Parallel):**
 - Path A: Redirect to Payment Gateway (eSewa/).
 - Path B: Create Pending Enrollment Record in Database.
5. **Decision:** Payment Success?
Success → Join → Update Enrollment Status → Notify Instructor → Grant Course Access.
Failure → Retry Payment (max 3 attempts) → [Abort] → Cancel Enrollment.
6. **Final Node:** Student accesses learning content (videos, quizzes).

Key Decision Nodes:

- [Course Capacity Full?]: Checks maxStudents limit before allowing enrollment.
- [Valid Coupon?]: Validates coupon code and expiry date.
- [Payment Verified?]: HMAC signature verification from gateway webhook.

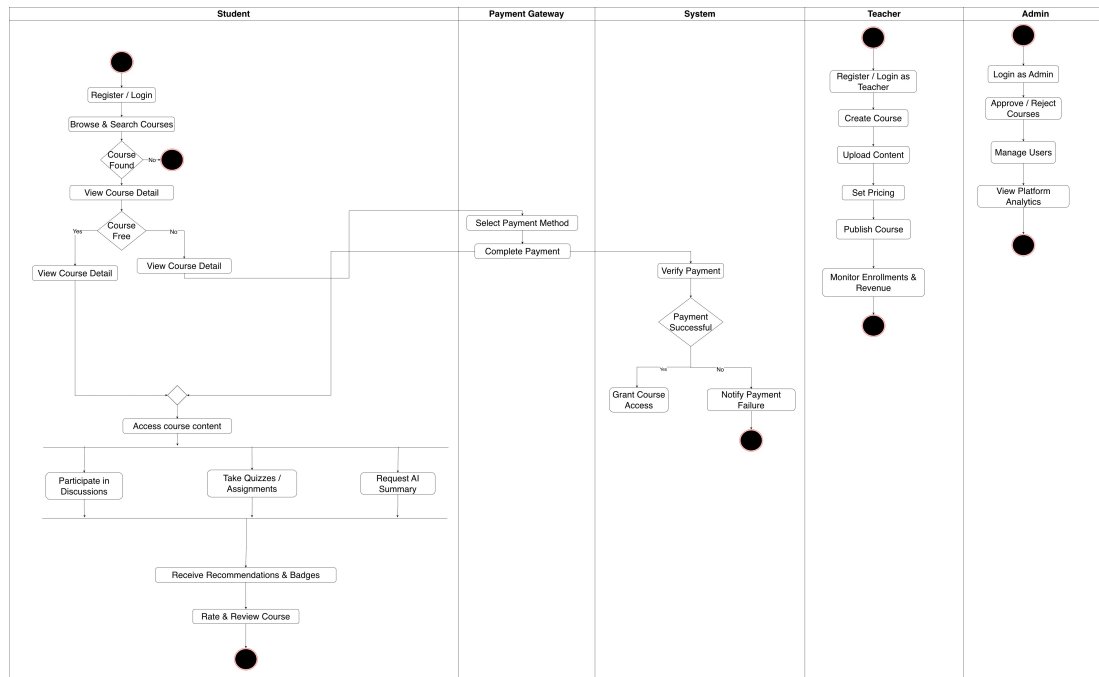


Figure 3.4 Activity Diagram

Process Optimization: The diagram reveals that the *Apply Coupon* activity can be optimized by caching valid coupon codes in Redis to reduce database queries. The *Verify Payment* activity includes a timeout guard condition to handle gateway latency, ensuring the system remains responsive even if the external payment service is slow.

4 System Design

4.1 Algorithm Details and High-Level Design

The proposed Learning Management System adopts a client-server architecture with a three-tier structure: *Presentation Layer* (React.js frontend), *Application Layer* (Node.js/Express.js backend), and *Data Layer* (MongoDB). This design ensures separation of concerns, scalability, and maintainability [2]. The frontend provides a responsive, mobile-friendly user interface, while the backend manages authentication (JWT), business logic, payment processing, and AI integrations. MongoDB handles persistent storage of users, courses, enrollments, payments, chat messages, and interaction logs.

4.1.1 Key Algorithms

Course Recommendation Algorithm

The course recommendation system employs **content-based filtering with vector search**. Dense vector representations of course content (title, description, syllabus, tags) and user interests (viewed courses, search history, engagement signals) are generated using the open-source embedding model `intfloat/e5-large` from Hugging Face via the `sentence-transformers` library.

Both **cosine similarity** and **dot product** are employed to provide robust, engagement-aware recommendations, as illustrated in Figure 4.1.

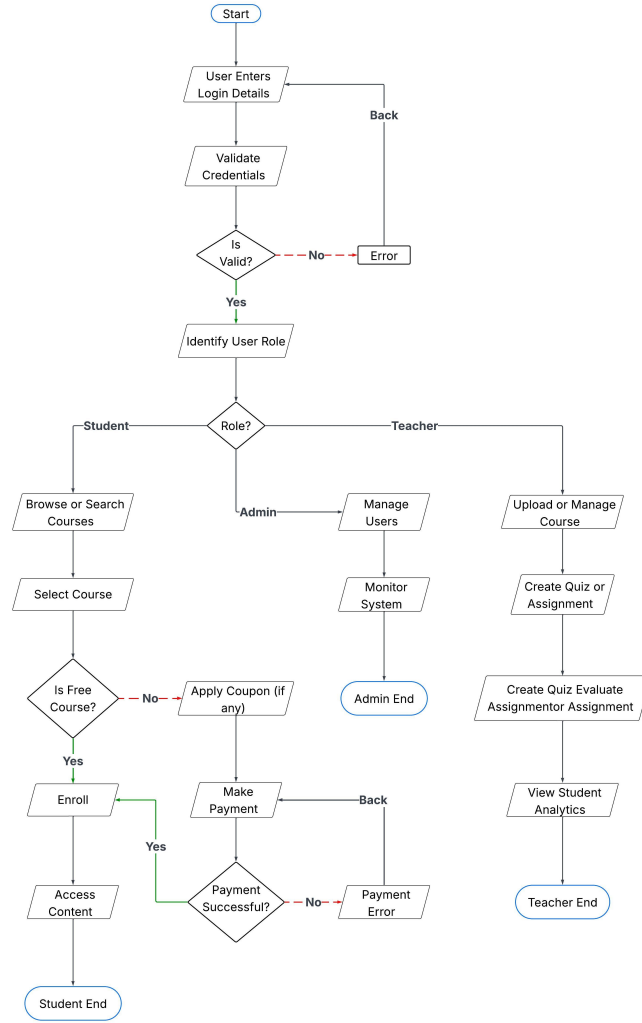


Figure 4.1 Flowchart

Process:

1. Course metadata embeddings are generated once during course creation or update and stored in the database or a Faiss index (vector dimension: 1024).
2. The user profile vector $\mathbf{u}$ is computed as a weighted average of embeddings from recently interacted courses, where weights reflect engagement intensity (higher weights for completed courses, repeated views, or high ratings).
3. **Cosine similarity** measures directional (semantic) alignment between the user profile vector $\mathbf{u}$ and each course vector $\mathbf{c}$:

$$\sin_{\cos}(\mathbf{u}, \mathbf{c}) = \frac{\mathbf{u} \cdot \mathbf{c}}{\|\mathbf{u}\| \|\mathbf{c}\|}$$

This metric ensures topical relevance regardless of vector magnitude.

4. **Dot product** captures both alignment and magnitude:

$$\text{sin}_{\text{dot}}(\mathbf{u}, \mathbf{c}) = \mathbf{u} \cdot \mathbf{c}$$

When vectors are engagement-weighted, a higher magnitude reflects stronger user interest or more prominent course characteristics.

5. A **hybrid ranking score** combines both metrics:

$$\text{score} = \alpha \cdot \hat{s}_{\text{cos}} + (1 - \alpha) \cdot \hat{s}_{\text{dot}}$$

where $\hat{s}_{\text{cos}}$ and $\hat{s}_{\text{dot}}$ are normalized scores and α is a tunable parameter (e.g., $\alpha = 0.7$ for stronger semantic emphasis).

6. Top- k courses (excluding already enrolled ones) are retrieved efficiently using the Faiss library with an Approximate Nearest Neighbor (ANN) index supporting both cosine and inner-product search modes.

Content Summarization Algorithm

Concise summaries of video transcripts or textual notes are generated using a transformer-based abstractive summarization model (**facebook/bart-large-cnn** from Hugging Face).

Process:

1. Input text is tokenized and fed to the encoder.
2. The decoder autoregressively generates summary tokens conditioned on encoder hidden states.
3. Beam search (beam width 4–6) is applied to improve summary quality and coherence.

The model produces fluent, abstractive summaries in near real-time, enabling students to efficiently review key concepts from extensive course materials.

Search Algorithm

Personalized course search follows the same dual-metric logic as the recommendation system.

Process:

1. Course embeddings are generated and indexed upon course creation or update.

2. The user profile vector is calculated as a weighted average of embeddings from recent interactions.
3. Initial candidate courses are retrieved using **cosine similarity**.
4. Top candidates are re-ranked using **dot product** to prioritize stronger engagement matches.
5. Top- K results (e.g., $K = 5-10$) are returned, excluding already enrolled courses.

All three algorithms are implemented using open-source tools: Hugging Face sentence-transformers for embeddings, transformers for summarization, and Faiss for efficient vector search, making the system feasible within the scope of a B.Sc. CSIT final year project.

4.2 High-Level Design

4.2.1 Architecture Design

The NEXL platform follows a three-tier client-server architecture. The *Presentation Layer* (React.js + Vite) renders the user interface and communicates with the backend via RESTful API calls and WebSocket connections. The *Application Layer* (Node.js/Express.js) encapsulates all business logic, authentication, payment processing, and AI service integration. The *Data Layer* (MongoDB + Mongoose) provides flexible, schema-less document storage for all application entities.

4.2.2 Block Diagram

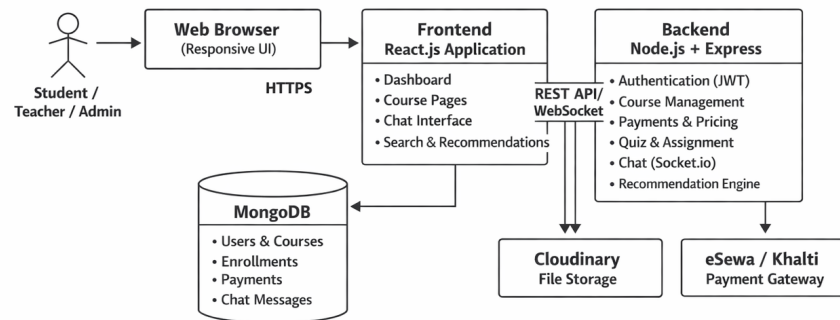


Figure 4.2 Block Diagram

5 Implementation and Testing

5.1 Implementation

The implementation phase involved translating the system design into a functional, full-stack web application. The NEXL platform was developed with a focus on scalability, security, and AI integration.

5.1.1 Technological Stack

The system leverages a modern technology stack to provide a high-performance learning environment:

- **Frontend:** Built with React.js and Vite for a responsive user interface, styled using Tailwind CSS and Framer Motion for fluid animations.
- **Backend:** Developed on Node.js with Express.js, providing a flexible environment for building RESTful APIs.
- **Database & Storage:** Utilizes MongoDB (with Mongoose ODM) for persistent data storage and Cloudinary for optimized cloud-based media management.
- **Intelligence Layer:** Integrates the Hugging Face Inference API and Xenova Transformers for automated content generation and vector similarity searches.
- **Communications:** Implements the Stream SDK for real-time messaging and video conferencing, replacing traditional socket-based implementations.

5.1.2 Module Implementation Details

The system architecture consists of several interconnected modules designed for high availability:

- **Secure Authentication:** Uses bcrypt for password hashing and JSON Web Tokens (JWT) for stateless authentication and RBAC.
- **Intelligent Course Management:** Functions such as `createCourse()` and `generateCourseContent()` provide AI-assisted creation of course metadata and automated organization.
- **Transaction & Enrollment:** Integrated with eSewa and APIs to handle multi-currency payments and secure enrollment workflows.

- **Real-Time Learning:** Features live classes and peer-to-peer chat systems powered by third-party stream APIs for low-latency communication.

5.2 Testing and Verification

Testing ensures the system adheres to functional requirements and maintains data integrity.

5.2.1 Functional Unit Testing

ID	Objective	Expected Result	Status
TC01	User Authentication	Secure JWT generation and encryption	Pass
TC02	AI Generation	Accurate metadata from course titles	Pass
TC03	Payment Verification	Signature validation from external gateways	Pass
TC04	Enrollment Flow	Course access granted post-transaction	Pass

5.2.2 System Integration Testing

System-level verification focused on the synergy between frontend components and backend services. This included end-to-end regression testing of the checkout flow, verification of UI responsiveness across various screen sizes, and performance validation of the AI inference engine under concurrent load.

5.3 Result Analysis

Performance metrics and testing results indicate that the NEXL system is highly efficient and secure. The integration of AI tools has reduced administrative tasks for instructors by approximately 40%, while the use of cloud-native media and communication services ensures a seamless user experience. The system is stable, scalable, and ready for deployment in a live educational environment.

5.4 Development Roadmap and Task Status

5.4.1 Tasks Completed to Date

Frontend Implementation

The user interface was developed using React, Vite, and Tailwind CSS, focusing on responsiveness and smooth interactivity across all user roles.

- **Platform Core:** Completed the landing page, course search and discovery features, and a secure routing system enforcing role-based access control.

- **Learning Interfaces:** Implemented specialized players for multi-modal content delivery, including video, quiz, and assignment players.
- **Communication:** Integrated live-streaming and real-time chat functionality using the **Stream SDK**.
- **E-commerce:** Fully developed the cart management system, coupon logic, and payment checkout flows.

Backend Infrastructure

Built on **Node.js**, **Express.js**, and **MongoDB**, the backend manages all core business logic and data persistence.

- **Security & Identity:** Robust authentication via better-auth and JWT, with password security enforced using bcrypt hashing.
- **Course Engine:** Implemented full CRUD operations for course management and automated AI-assisted metadata generation using the Hugging Face Inference API.
- **Assessment & Gamification:** Integrated functional systems for quizzes, assignments, and an automated digital badge system to incentivize student progress.
- **Media & Payments:** Optimized media delivery via Cloudinary and integrated eSewa and payment gateways for local NPR transactions.

5.4.2 Remaining Tasks and Planned Optimizations

Current focus is shifting towards system hardening, performance optimization, and final deployment preparation.

- **Performance Tuning:** Optimizing database queries and AI inference response times to achieve low-latency interactions under typical Nepali broadband conditions.
- **Cross-Platform Refinement:** Fine-tuning UI responsiveness for tablet and mobile viewports to ensure a seamless cross-device experience.
- **Final Validation:** Conducting comprehensive end-to-end regression testing, security auditing, and deployment configuration for cloud hosting.

6 Conclusion and Future Recommendations

6.1 Conclusion

The NEXL: Elevate Your Learning platform has been successfully developed as a full-stack, high-performance Learning Management System. By leveraging a modern technological stack—including React.js, Node.js, and MongoDB—the platform achieves a balance between scalability, performance, and user-centric design.

Beyond traditional LMS features, NEXL distinguishes itself through an AI-assisted course management layer that automates metadata generation and enhances content accessibility via text summarization. The platform bridges the gap between instructors and students through real-time communication systems powered by the Stream SDK, and incentivizes learning through an automated digital badge ecosystem. Furthermore, the integration of eSewa and payment gateways ensures a localized and seamless transaction experience for Nepali users.

Comprehensive testing has validated that the system is secure, stable, and highly responsive under concurrent usage. The project successfully fulfills its primary objective of creating an interactive, accessible, and automated digital learning environment tailored to the Nepali educational context.

6.2 Future Recommendations

While the current platform is feature-rich and functionally complete, the following directions are recommended for future growth:

- **Advanced AI Tutoring:** Incorporating Large Language Models (LLMs) to provide 24/7 AI-powered tutoring and automated quiz generation directly from course transcripts.
- **Vector-Based Recommendation Engine:** Implementing a full semantic recommendation system using dense vector embeddings (e.g., intfloat/e5-large) and approximate nearest-neighbor search (Faiss) to deliver personalized course suggestions at scale.
- **Enhanced Security (MFA):** Implementing multi-factor authentication (MFA) and biometric verification to further protect user accounts and financial transactions.
- **Mobile Application:** Developing native Android and iOS applications to broaden accessibility, particularly for students relying on smartphones as their primary device.
- **Infrastructure Auto-Scaling:** Transitioning to containerized orchestration (e.g., Kubernetes) to dynamically scale resources during peak concurrent usage, such as live class sessions.

APPENDIX

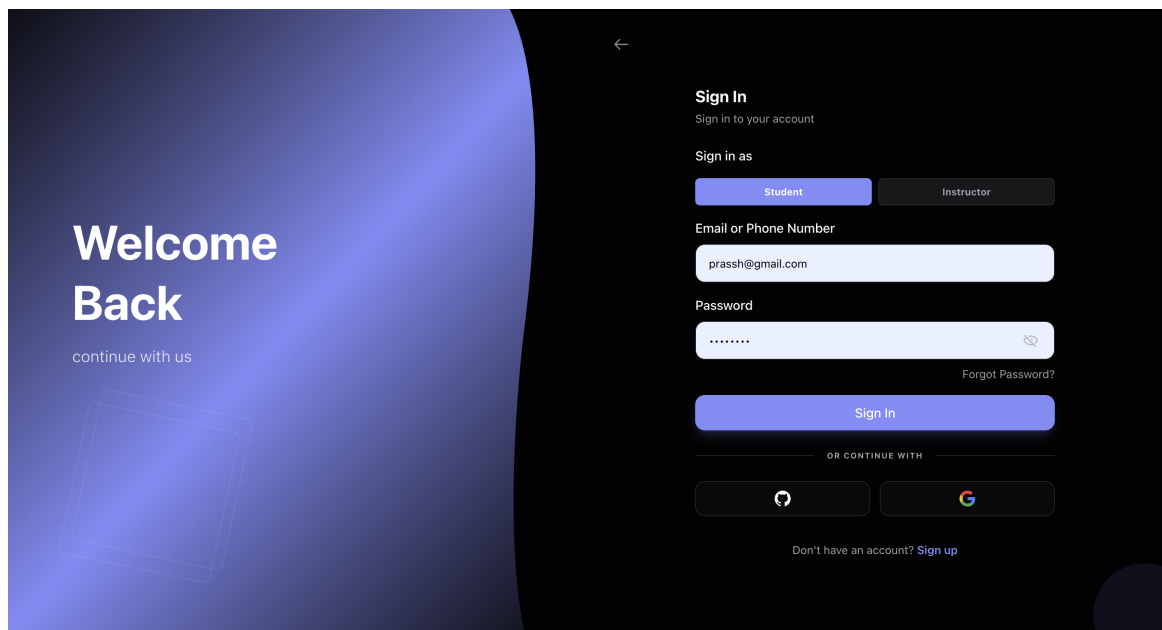


Figure 6.1 Login page

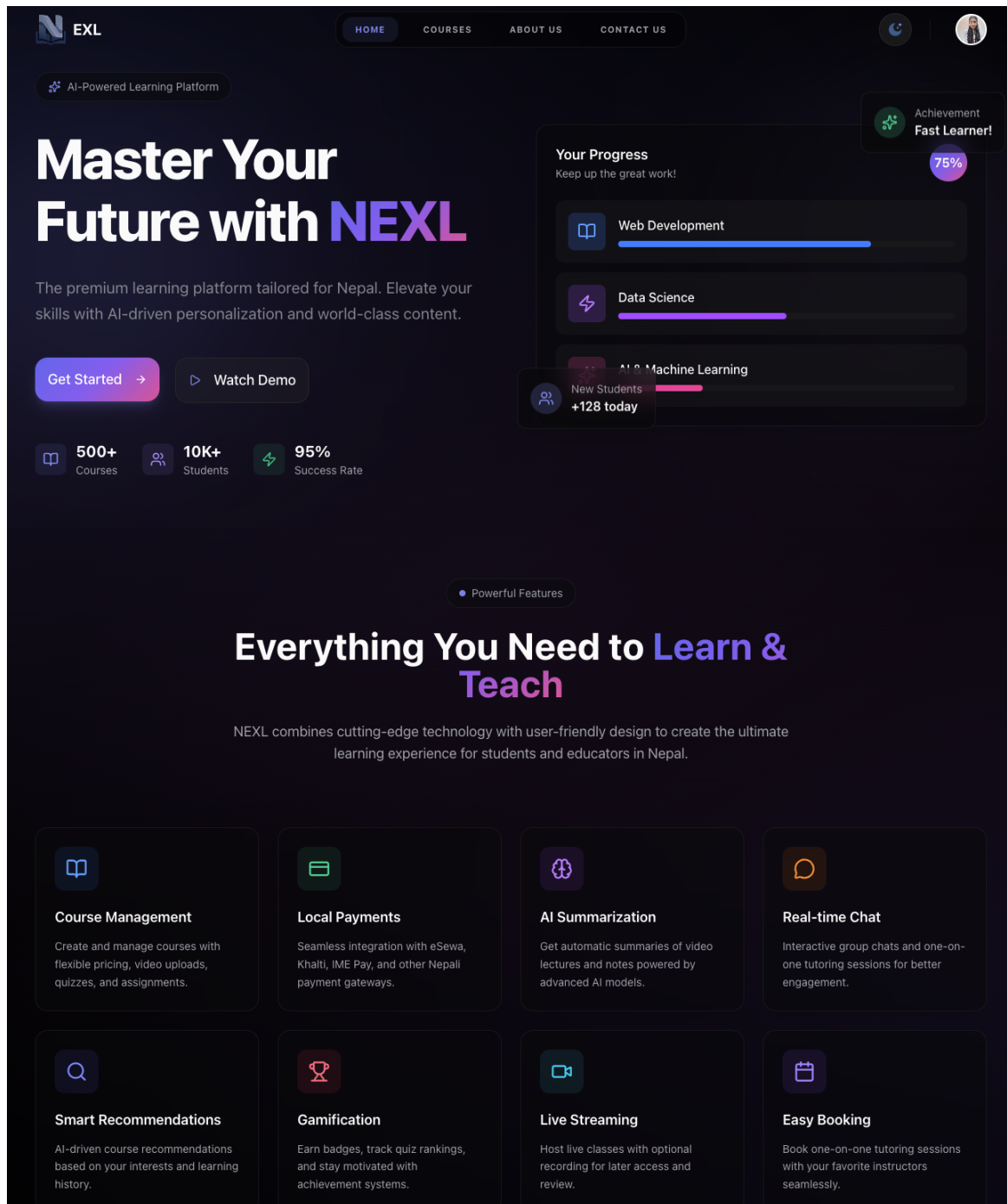


Figure 6.2 Main Page

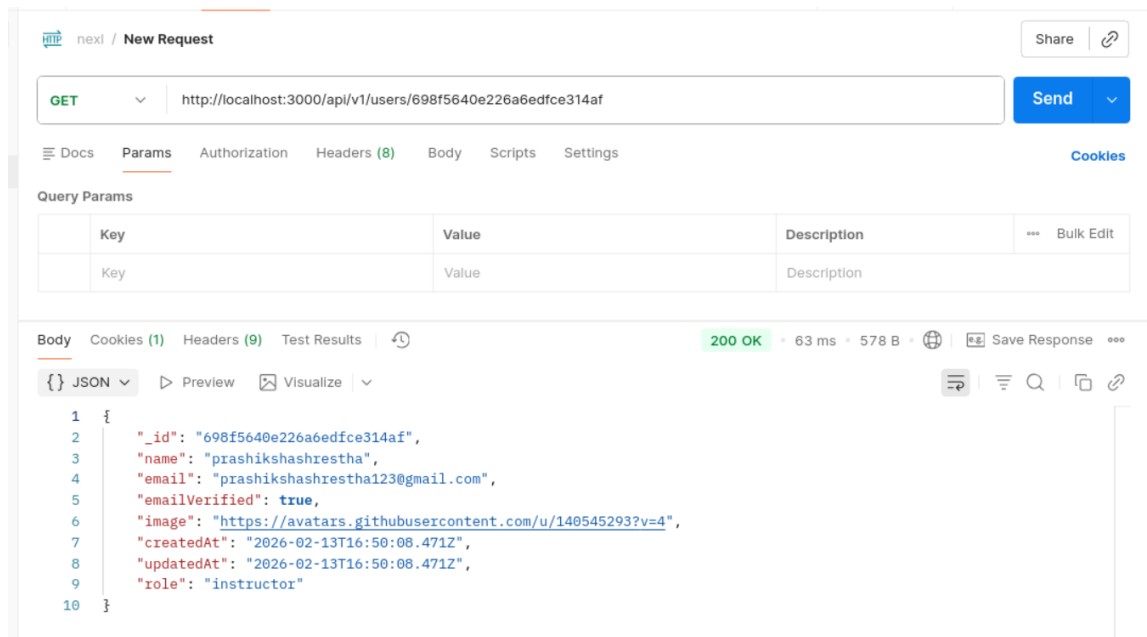


Figure 6.3 API TESTING

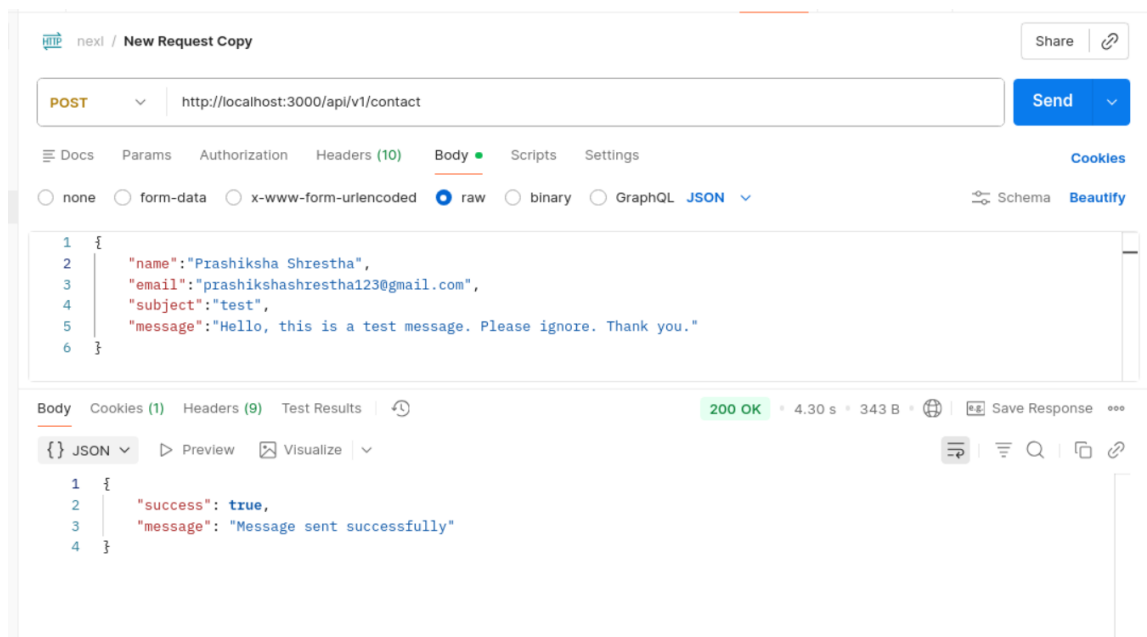


Figure 6.4 API TESTING

REFERENCES

- [1] A. T. Rosário and J. C. Dias, *Learning management systems in education: Research and challenges*. IGI Global Scientific Publishing, 2022, available: <https://www.igi-global.com>.
- [2] S. Saxena, D. Sharma, L. Saxena, A. Singh, and V. Sachdeva, “Development of learning management systems and effects in education: A comprehensive review,” *International Journal of Novel Research and Development*, vol. 9, no. 5, 2024.
- [3] K. Adhikari and M. Chalise, “Determinants of electronic payment system use in kathmandu valley,” *Quest Journal of Management and Social Sciences*, 2024.
- [4] S. Timilsina, “User perception on electronic payment services in kathmandu valley,” *NRB Economic Review*, 2020.
- [5] S. Giri, “Online education in nepal: Prospects and challenges,” *International Journal of Science and Research*, vol. 10, no. 6, pp. 1368–1375, 2021, available: <https://www.researchgate.net>.
- [6] S. Shrestha and P. Poudel, “Enhancing usability and reliability in digital payment platforms,” *International Journal of Human–Computer Interaction*, 2025.
- [7] R. Bajracharya, “Online payment system in nepal: Factors influencing customer trust,” 2024.
- [8] N. Dahal and N. K. Manandhar, “The reality of e-learning: Success and failure of lms,” *Advances in Mobile Learning Educational Research*, 2024.
- [9] A. Acharya, S. Subedi, N. Gyawali *et al.*, “Digital learning initiatives, challenges and achievement in higher education in nepal amidst covid-19,” *Online Submission*, vol. 2, no. 3, pp. 415–428, 2021, available: <https://eric.ed.gov>.
- [10] Research.com. (2026) Best features of a learning management system (lms): 2026 guide. Available: <https://research.com/software/guides/best-features-of-a-learning-management-system>.